

## Task 3– IPL SQL Analysis: 50 Questions

### Part A — Basic Queries and Filtering

1. Display the first 10 records from the matches table.

The screenshot shows a SQL IDE interface. The query editor contains the following SQL code:

```
1 • CREATE DATABASE ipl_project;
2 • USE ipl_project;
3 • SET GLOBAL local_infile = 1;
4
5 • SELECT * from matches
6 • LIMIT 10;
```

Below the query editor, the 'Result Grid' tab is active, displaying the first 10 records from the matches table. The table has the following columns: id, season, city, date, team1, team2, toss\_winner, and toss\_de. The data is as follows:

| id | season | city      | date       | team1                       | team2                       | toss_winner                 | toss_de |
|----|--------|-----------|------------|-----------------------------|-----------------------------|-----------------------------|---------|
| 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad         | Royal Challengers Bangalore | Royal Challengers Bangalore | field   |
| 2  | 2017   | Pune      | 2017-04-06 | Mumbai Indians              | Rising Pune Supergiant      | Rising Pune Supergiant      | field   |
| 3  | 2017   | Rajkot    | 2017-04-07 | Gujarat Lions               | Kolkata Knight Riders       | Kolkata Knight Riders       | field   |
| 4  | 2017   | Indore    | 2017-04-08 | Rising Pune Supergiant      | Kings XI Punjab             | Kings XI Punjab             | field   |
| 5  | 2017   | Bangalore | 2017-04-08 | Royal Challengers Bangalore | Delhi Daredevils            | Royal Challengers Bangalore | bat     |
| 6  | 2017   | Hyderabad | 2017-04-09 | Gujarat Lions               | Sunrisers Hyderabad         | Sunrisers Hyderabad         | field   |
| 7  | 2017   | Mumbai    | 2017-04-09 | Kolkata Knight Riders       | Mumbai Indians              | Mumbai Indians              | field   |
| 8  | 2017   | Indore    | 2017-04-10 | Royal Challengers Bangalore | Kings XI Punjab             | Royal Challengers Bangalore | bat     |
| 9  | 2017   | Pune      | 2017-04-11 | Delhi Daredevils            | Rising Pune Supergiant      | Rising Pune Supergiant      | field   |
| 10 | 2017   | Mumbai    | 2017-04-12 | Sunrisers Hyderabad         | Mumbai Indians              | Mumbai Indians              | field   |

2. Display the first 10 records from the deliveries table.

The screenshot shows a SQL IDE interface. The query editor contains the following SQL code:

```
3 • SET GLOBAL local_infile = 1;
4
5 • SELECT * from matches
6 • LIMIT 10;
7
8 • SELECT * from deliveries
9 • LIMIT 10;
```

Below the query editor, the 'Result Grid' tab is active, displaying the first 10 records from the deliveries table. The table has the following columns: match\_id, inning, batting\_team, bowling\_team, over, ball, batsman, non\_striker, bowler, and is\_super\_ov. The data is as follows:

| match_id | inning | batting_team        | bowling_team                | over | ball | batsman   | non_striker | bowler      | is_super_ov |
|----------|--------|---------------------|-----------------------------|------|------|-----------|-------------|-------------|-------------|
| 1        | 1      | Sunrisers Hyderabad | Royal Challengers Bangalore | 1    | 1    | DA Warner | S Dhawan    | TS Mills    | 0           |
| 1        | 1      | Sunrisers Hyderabad | Royal Challengers Bangalore | 1    | 2    | DA Warner | S Dhawan    | TS Mills    | 0           |
| 1        | 1      | Sunrisers Hyderabad | Royal Challengers Bangalore | 1    | 3    | DA Warner | S Dhawan    | TS Mills    | 0           |
| 1        | 1      | Sunrisers Hyderabad | Royal Challengers Bangalore | 1    | 4    | DA Warner | S Dhawan    | TS Mills    | 0           |
| 1        | 1      | Sunrisers Hyderabad | Royal Challengers Bangalore | 1    | 5    | DA Warner | S Dhawan    | TS Mills    | 0           |
| 1        | 1      | Sunrisers Hyderabad | Royal Challengers Bangalore | 1    | 6    | S Dhawan  | DA Warner   | TS Mills    | 0           |
| 1        | 1      | Sunrisers Hyderabad | Royal Challengers Bangalore | 1    | 7    | S Dhawan  | DA Warner   | TS Mills    | 0           |
| 1        | 1      | Sunrisers Hyderabad | Royal Challengers Bangalore | 2    | 1    | S Dhawan  | DA Warner   | A Choudhary | 0           |
| 1        | 1      | Sunrisers Hyderabad | Royal Challengers Bangalore | 2    | 2    | DA Warner | S Dhawan    | A Choudhary | 0           |
| 1        | 1      | Sunrisers Hyderabad | Royal Challengers Bangalore | 2    | 3    | DA Warner | S Dhawan    | A Choudhary | 0           |

3. Count the total number of rows in the matches table.

The screenshot shows a database query editor interface. The SQL query is as follows:

```
6  LIMIT 10;  
7  
8  • SELECT * from deliveries  
9  LIMIT 10;  
10  
11 • SELECT COUNT(*) from matches;  
12
```

Below the query editor, the 'Result Grid' is displayed with the following data:

| COUNT(*) |
|----------|
| 756      |

The interface includes a toolbar at the top with various icons and a 'Limit to 1000 rows' dropdown. On the right side, there is a vertical toolbar with buttons for 'Result Grid', 'Form Editor', and 'Field Types'. At the bottom, a tab labeled 'Result 3' is visible, and a 'Read Only' status is indicated.

4. Count the total number of rows in the deliveries table.

The screenshot shows the same database query editor interface. The SQL query is as follows:

```
8  • SELECT * from deliveries  
9  LIMIT 10;  
10  
11 • SELECT COUNT(*) from matches;  
12  
13 • SELECT COUNT(*) from deliveries;  
14
```

Below the query editor, the 'Result Grid' is displayed with the following data:

| COUNT(*) |
|----------|
| 60889    |

The interface includes a toolbar at the top with various icons and a 'Limit to 1000 rows' dropdown. On the right side, there is a vertical toolbar with buttons for 'Result Grid', 'Form Editor', and 'Field Types'. At the bottom, a tab labeled 'Result 6' is visible, and a 'Read Only' status is indicated.

5. Display all unique IPL seasons from the matches table.

```
10
11 • SELECT COUNT(*) from matches;
12
13 • SELECT COUNT(*) from deliveries;
14
15 • SELECT distinct season
16   from matches;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

| season |
|--------|
| 2017   |
| 2008   |
| 2009   |
| 2010   |
| 2011   |
| 2012   |
| 2013   |
| 2014   |
| 2015   |
| 2016   |
| 2018   |
| 2019   |

matches 7 x | Read Only

6. Display all unique cities where IPL matches were played

```
13 • SELECT COUNT(*) from deliveries;
14
15 • SELECT distinct season
16   from matches;
17
18 • SELECT distinct city
19   from matches;
```

Limit to 1000 rows |

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

| city       |
|------------|
| Hyderabad  |
| Pune       |
| Rajkot     |
| Indore     |
| Bangalore  |
| Mumbai     |
| Kolkata    |
| Delhi      |
| Chandigarh |
| Kanpur     |
| Jaipur     |
| Chennai    |

7. Display all unique team names appearing in the team1 column.

The screenshot shows a SQL IDE interface. The query editor contains the following SQL code:

```
16 from matches;
17
18 • SELECT distinct city
19 from matches;
20
21 • SELECT distinct team1
22 from matches;
```

The query is executed, and the Result Grid displays the following data:

| team1                       |
|-----------------------------|
| Sunrisers Hyderabad         |
| Mumbai Indians              |
| Gujarat Lions               |
| Rising Pune Supergiant      |
| Royal Challengers Bangalore |
| Kolkata Knight Riders       |
| Delhi Daredevils            |
| Kings XI Punjab             |
| Chennai Super Kings         |
| Rajasthan Royals            |
| Deccan Chargers             |
| Kochi Tuskers Kerala        |

8. Display all matches played in the 2017 season.

The screenshot shows a SQL IDE interface. The query editor contains the following SQL code:

```
19 from matches;
20
21 • SELECT distinct team1
22 from matches;
23
24 • SELECT * from matches
25 where season = 2017;
```

The query is executed, and the Result Grid displays the following data:

|   | id | season | city      | date       | team1                       | team2                       | toss_winner                 | toss  |
|---|----|--------|-----------|------------|-----------------------------|-----------------------------|-----------------------------|-------|
| ▶ | 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad         | Royal Challengers Bangalore | Royal Challengers Bangalore | field |
|   | 2  | 2017   | Pune      | 2017-04-06 | Mumbai Indians              | Rising Pune Supergiant      | Rising Pune Supergiant      | field |
|   | 3  | 2017   | Rajkot    | 2017-04-07 | Gujarat Lions               | Kolkata Knight Riders       | Kolkata Knight Riders       | field |
|   | 4  | 2017   | Indore    | 2017-04-08 | Rising Pune Supergiant      | Kings XI Punjab             | Kings XI Punjab             | field |
|   | 5  | 2017   | Bangalore | 2017-04-08 | Royal Challengers Bangalore | Delhi Daredevils            | Royal Challengers Bangalore | bat   |
|   | 6  | 2017   | Hyderabad | 2017-04-09 | Gujarat Lions               | Sunrisers Hyderabad         | Sunrisers Hyderabad         | field |
|   | 7  | 2017   | Mumbai    | 2017-04-09 | Kolkata Knight Riders       | Mumbai Indians              | Mumbai Indians              | field |
|   | 8  | 2017   | Indore    | 2017-04-10 | Royal Challengers Bangalore | Kings XI Punjab             | Royal Challengers Bangalore | bat   |
|   | 9  | 2017   | Pune      | 2017-04-11 | Delhi Daredevils            | Rising Pune Supergiant      | Rising Pune Supergiant      | field |
|   | 10 | 2017   | Mumbai    | 2017-04-12 | Sunrisers Hyderabad         | Mumbai Indians              | Mumbai Indians              | field |
|   | 11 | 2017   | Kolkata   | 2017-04-13 | Kings XI Punjab             | Kolkata Knight Riders       | Kolkata Knight Riders       | field |

9. Display all matches where the toss decision was field.

The screenshot shows a database query editor with a toolbar at the top. The SQL query is as follows:

```
22 from matches;  
23  
24 • SELECT * from matches  
25 where season = 2017;  
26  
27 • SELECT * from matches  
28 where toss_decision = 'field';
```

Below the query editor is a "Result Grid" section. It includes a "Filter Rows:" input field, an "Export:" button, and a "Wrap Cell Content:" checkbox. The grid displays the following data:

|  | id | season | city      | date       | team1                       | team2                       | toss_winner                 | toss  |
|--|----|--------|-----------|------------|-----------------------------|-----------------------------|-----------------------------|-------|
|  | 11 | 2017   | Kolkata   | 2017-04-13 | Kings XI Punjab             | Kolkata Knight Riders       | Kolkata Knight Riders       | field |
|  | 12 | 2017   | Bangalore | 2017-04-14 | Royal Challengers Bangalore | Mumbai Indians              | Mumbai Indians              | field |
|  | 13 | 2017   | Rajkot    | 2017-04-14 | Rising Pune Supergiant      | Gujarat Lions               | Gujarat Lions               | field |
|  | 14 | 2017   | Kolkata   | 2017-04-15 | Kolkata Knight Riders       | Sunrisers Hyderabad         | Sunrisers Hyderabad         | field |
|  | 16 | 2017   | Mumbai    | 2017-04-16 | Gujarat Lions               | Mumbai Indians              | Mumbai Indians              | field |
|  | 17 | 2017   | Bangalore | 2017-04-16 | Rising Pune Supergiant      | Royal Challengers Bangalore | Royal Challengers Bangalore | field |
|  | 19 | 2017   | Hyderabad | 2017-04-17 | Sunrisers Hyderabad         | Kings XI Punjab             | Kings XI Punjab             | field |
|  | 20 | 2017   | Rajkot    | 2017-04-18 | Royal Challengers Bangalore | Gujarat Lions               | Gujarat Lions               | field |
|  | 22 | 2017   | Indore    | 2017-04-20 | Kings XI Punjab             | Mumbai Indians              | Mumbai Indians              | field |
|  | 23 | 2017   | Kolkata   | 2017-04-21 | Kolkata Knight Riders       | Gujarat Lions               | Gujarat Lions               | field |
|  | 24 | 2017   | Mumbai    | 2017-04-22 | Mumbai Indians              | Delhi Daredevils            | Delhi Daredevils            | field |

On the right side of the interface, there are buttons for "Result Grid", "Form Editor", and "Field Types".

10. Display all matches won by Mumbai Indians

The screenshot shows a database query editor with a toolbar at the top. The SQL query is as follows:

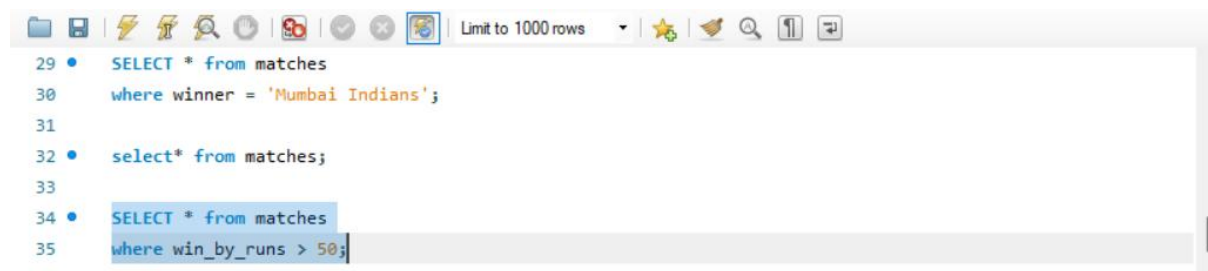
```
25 where season = 2017;  
26  
27 • SELECT * from matches  
28 where toss_decision = 'field';  
29  
30 • SELECT * from matches  
31 where winner = 'Mumbai Indians';
```

Below the query editor is a "Result Grid" section. It includes a "Filter Rows:" input field, an "Export:" button, and a "Wrap Cell Content:" checkbox. The grid displays the following data:

|   | id | season | city      | date       | team1                       | team2                 | toss_winner                 | toss_decision |
|---|----|--------|-----------|------------|-----------------------------|-----------------------|-----------------------------|---------------|
| ▶ | 7  | 2017   | Mumbai    | 2017-04-09 | Kolkata Knight Riders       | Mumbai Indians        | Mumbai Indians              | field         |
|   | 10 | 2017   | Mumbai    | 2017-04-12 | Sunrisers Hyderabad         | Mumbai Indians        | Mumbai Indians              | field         |
|   | 12 | 2017   | Bangalore | 2017-04-14 | Royal Challengers Bangalore | Mumbai Indians        | Mumbai Indians              | field         |
|   | 16 | 2017   | Mumbai    | 2017-04-16 | Gujarat Lions               | Mumbai Indians        | Mumbai Indians              | field         |
|   | 22 | 2017   | Indore    | 2017-04-20 | Kings XI Punjab             | Mumbai Indians        | Mumbai Indians              | field         |
|   | 24 | 2017   | Mumbai    | 2017-04-22 | Mumbai Indians              | Delhi Daredevils      | Delhi Daredevils            | field         |
|   | 34 | 2017   | Rajkot    | 2017-04-29 | Gujarat Lions               | Mumbai Indians        | Gujarat Lions               | bat           |
|   | 37 | 2017   | Mumbai    | 2017-05-01 | Royal Challengers Bangalore | Mumbai Indians        | Royal Challengers Bangalore | bat           |
|   | 44 | 2017   | Delhi     | 2017-05-06 | Mumbai Indians              | Delhi Daredevils      | Delhi Daredevils            | field         |
|   | 53 | 2017   | Kolkata   | 2017-05-13 | Mumbai Indians              | Kolkata Knight Riders | Kolkata Knight Riders       | field         |
|   | 58 | 2017   | Bangalore | 2017-05-19 | Kolkata Knight Riders       | Mumbai Indians        | Mumbai Indians              | field         |

On the right side of the interface, there are buttons for "Result Grid", "Form Editor", and "Field Types".

11. Display matches where win\_by\_runs is greater than 50.



```

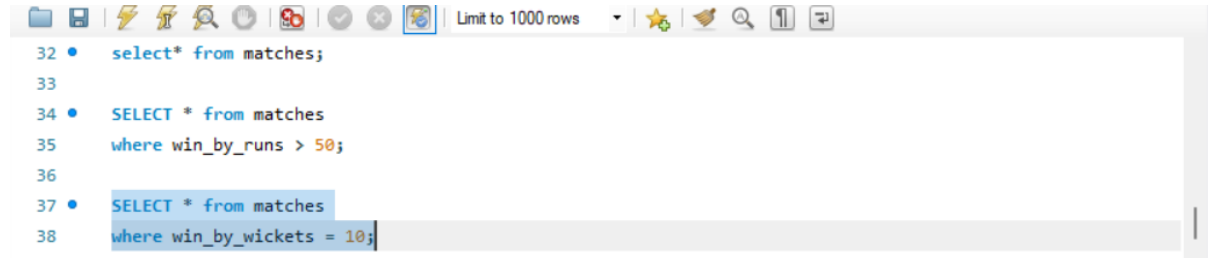
29 • SELECT * from matches
30   where winner = 'Mumbai Indians';
31
32 • select* from matches;
33
34 • SELECT * from matches
35   where win_by_runs > 50;

```

Result Grid

|   | id  | season | city           | date       | team1                       | team2                       | toss_winner                 | toss  |
|---|-----|--------|----------------|------------|-----------------------------|-----------------------------|-----------------------------|-------|
| ▶ | 9   | 2017   | Pune           | 2017-04-11 | Delhi Daredevils            | Rising Pune Supergiant      | Rising Pune Supergiant      | field |
|   | 15  | 2017   | Delhi          | 2017-04-15 | Delhi Daredevils            | Kings XI Punjab             | Delhi Daredevils            | bat   |
|   | 27  | 2017   | Kolkata        | 2017-04-23 | Kolkata Knight Riders       | Royal Challengers Bangalore | Royal Challengers Bangalore | field |
|   | 33  | 2017   | Pune           | 2017-04-29 | Rising Pune Supergiant      | Royal Challengers Bangalore | Royal Challengers Bangalore | field |
|   | 44  | 2017   | Delhi          | 2017-05-06 | Mumbai Indians              | Delhi Daredevils            | Delhi Daredevils            | field |
|   | 60  | 2008   | Bangalore      | 2008-04-18 | Kolkata Knight Riders       | Royal Challengers Bangalore | Royal Challengers Bangalore | field |
|   | 69  | 2008   | Chandigarh     | 2008-04-25 | Kings XI Punjab             | Mumbai Indians              | Mumbai Indians              | field |
|   | 101 | 2008   | Jaipur         | 2008-05-17 | Rajasthan Royals            | Royal Challengers Bangalore | Royal Challengers Bangalore | field |
|   | 115 | 2008   | Mumbai         | 2008-05-30 | Rajasthan Royals            | Delhi Daredevils            | Delhi Daredevils            | field |
|   | 119 | 2009   | Cape Town      | 2009-04-18 | Royal Challengers Bangalore | Rajasthan Royals            | Royal Challengers Bangalore | bat   |
|   | 122 | 2009   | Port Elizab... | 2009-04-20 | Chennai Super Kings         | Royal Challengers Bangalore | Chennai Super Kings         | bat   |

12. Display matches where win\_by\_wickets equals 10.



```

32 • select* from matches;
33
34 • SELECT * from matches
35   where win_by_runs > 50;
36
37 • SELECT * from matches
38   where win_by_wickets = 10;

```

Result Grid

|   | id   | season | city       | date       | team1            | team2                       | toss_winner                 | toss_decision | re |
|---|------|--------|------------|------------|------------------|-----------------------------|-----------------------------|---------------|----|
| ▶ | 3    | 2017   | Rajkot     | 2017-04-07 | Gujarat Lions    | Kolkata Knight Riders       | Kolkata Knight Riders       | field         | no |
|   | 35   | 2017   | Chandigarh | 2017-04-30 | Delhi Daredevils | Kings XI Punjab             | Kings XI Punjab             | field         | no |
|   | 72   | 2008   | Mumbai     | 2008-04-27 | Mumbai Indians   | Deccan Chargers             | Deccan Chargers             | field         | no |
|   | 120  | 2009   | Cape Town  | 2009-04-19 | Kings XI Punjab  | Delhi Daredevils            | Delhi Daredevils            | field         | no |
|   | 184  | 2010   | Bangalore  | 2010-03-18 | Rajasthan Royals | Royal Challengers Bangalore | Royal Challengers Bangalore | field         | no |
|   | 299  | 2011   | Mumbai     | 2011-05-20 | Mumbai Indians   | Rajasthan Royals            | Mumbai Indians              | bat           | no |
|   | 377  | 2012   | Jaipur     | 2012-05-20 | Rajasthan Royals | Mumbai Indians              | Rajasthan Royals            | bat           | no |
|   | 391  | 2013   | Chandigarh | 2013-04-10 | Kings XI Punjab  | Chennai Super Kings         | Chennai Super Kings         | field         | no |
|   | 543  | 2015   | Delhi      | 2015-04-26 | Delhi Daredevils | Royal Challengers Bangalore | Royal Challengers Bangalore | field         | no |
|   | 591  | 2016   | Rajkot     | 2016-04-21 | Gujarat Lions    | Sunrisers Hyderabad         | Sunrisers Hyderabad         | field         | no |
|   | 7941 | 2018   | Indore     | 14/05/18   | Kings XI Punjab  | Royal Challengers Bangalore | Royal Challengers Bangalore | field         | no |

13. Display matches where the toss winner and match winner are the same team.

SQL Editor interface showing a query and its results.

```

35 where win_by_runs > 50;
36
37 • SELECT * from matches
38 where win_by_wickets = 10;
39
40 • SELECT * from matches
41 where toss_winner = winner;

```

Result Grid

|   | id | season | city      | date       | team1                       | team2                  | toss_winner                 | toss_decisi |
|---|----|--------|-----------|------------|-----------------------------|------------------------|-----------------------------|-------------|
| ▶ | 2  | 2017   | Pune      | 2017-04-06 | Mumbai Indians              | Rising Pune Supergiant | Rising Pune Supergiant      | field       |
|   | 3  | 2017   | Rajkot    | 2017-04-07 | Gujarat Lions               | Kolkata Knight Riders  | Kolkata Knight Riders       | field       |
|   | 4  | 2017   | Indore    | 2017-04-08 | Rising Pune Supergiant      | Kings XI Punjab        | Kings XI Punjab             | field       |
|   | 5  | 2017   | Bangalore | 2017-04-08 | Royal Challengers Bangalore | Delhi Daredevils       | Royal Challengers Bangalore | bat         |
|   | 6  | 2017   | Hyderabad | 2017-04-09 | Gujarat Lions               | Sunrisers Hyderabad    | Sunrisers Hyderabad         | field       |
|   | 7  | 2017   | Mumbai    | 2017-04-09 | Kolkata Knight Riders       | Mumbai Indians         | Mumbai Indians              | field       |
|   | 10 | 2017   | Mumbai    | 2017-04-12 | Sunrisers Hyderabad         | Mumbai Indians         | Mumbai Indians              | field       |
|   | 11 | 2017   | Kolkata   | 2017-04-13 | Kings XI Punjab             | Kolkata Knight Riders  | Kolkata Knight Riders       | field       |
|   | 12 | 2017   | Bangalore | 2017-04-14 | Royal Challengers Bangalore | Mumbai Indians         | Mumbai Indians              | field       |
|   | 13 | 2017   | Rajkot    | 2017-04-14 | Rising Pune Supergiant      | Gujarat Lions          | Gujarat Lions               | field       |
|   | 15 | 2017   | Delhi     | 2017-04-15 | Delhi Daredevils            | Kings XI Punjab        | Delhi Daredevils            | bat         |

14. Display matches where Mumbai Indians appears as either team1 or team2.

SQL Editor interface showing a query and its results.

```

38 where win_by_wickets = 10;
39
40 • SELECT * from matches
41 where toss_winner = winner;
42
43 • SELECT * from matches
44 where team1 = 'Mumbai Indians' OR team2 = 'Mumbai Indians';

```

Result Grid

|   | id | season | city      | date       | team1                       | team2                  | toss_winner                 | toss_decisi |
|---|----|--------|-----------|------------|-----------------------------|------------------------|-----------------------------|-------------|
| ▶ | 2  | 2017   | Pune      | 2017-04-06 | Mumbai Indians              | Rising Pune Supergiant | Rising Pune Supergiant      | field       |
|   | 7  | 2017   | Mumbai    | 2017-04-09 | Kolkata Knight Riders       | Mumbai Indians         | Mumbai Indians              | field       |
|   | 10 | 2017   | Mumbai    | 2017-04-12 | Sunrisers Hyderabad         | Mumbai Indians         | Mumbai Indians              | field       |
|   | 12 | 2017   | Bangalore | 2017-04-14 | Royal Challengers Bangalore | Mumbai Indians         | Mumbai Indians              | field       |
|   | 16 | 2017   | Mumbai    | 2017-04-16 | Gujarat Lions               | Mumbai Indians         | Mumbai Indians              | field       |
|   | 22 | 2017   | Indore    | 2017-04-20 | Kings XI Punjab             | Mumbai Indians         | Mumbai Indians              | field       |
|   | 24 | 2017   | Mumbai    | 2017-04-22 | Mumbai Indians              | Delhi Daredevils       | Delhi Daredevils            | field       |
|   | 28 | 2017   | Mumbai    | 2017-04-24 | Rising Pune Supergiant      | Mumbai Indians         | Mumbai Indians              | field       |
|   | 34 | 2017   | Rajkot    | 2017-04-29 | Gujarat Lions               | Mumbai Indians         | Gujarat Lions               | bat         |
|   | 37 | 2017   | Mumbai    | 2017-05-01 | Royal Challengers Bangalore | Mumbai Indians         | Royal Challengers Bangalore | bat         |
|   | 44 | 2017   | Delhi     | 2017-05-06 | Mumbai Indians              | Delhi Daredevils       | Delhi Daredevils            | field       |

matches 18 × Read Only

15. Display the 10 matches with the highest win\_by\_runs values.

Limit to 1000 rows

```

42
43 • SELECT * from matches
44 where team1 = 'Mumbai Indians' OR team2 = 'Mumbai Indians';
45
46 • SELECT * from matches
47 order by win_by_runs DESC
48 limit 10;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows: |

|   | id    | season | city       | date       | team1                       | team2                       | toss_winner                 | toss_c |
|---|-------|--------|------------|------------|-----------------------------|-----------------------------|-----------------------------|--------|
| ▶ | 44    | 2017   | Delhi      | 2017-05-06 | Mumbai Indians              | Delhi Daredevils            | Delhi Daredevils            | field  |
|   | 620   | 2016   | Bangalore  | 2016-05-14 | Royal Challengers Bangalore | Gujarat Lions               | Gujarat Lions               | field  |
|   | 60    | 2008   | Bangalore  | 2008-04-18 | Kolkata Knight Riders       | Royal Challengers Bangalore | Royal Challengers Bangalore | field  |
|   | 557   | 2015   | Bangalore  | 2015-05-06 | Royal Challengers Bangalore | Kings XI Punjab             | Kings XI Punjab             | field  |
|   | 411   | 2013   | Bangalore  | 2013-04-23 | Royal Challengers Bangalore | Pune Warriors               | Pune Warriors               | field  |
|   | 11147 | 2019   | Hyderabad  | 31/03/19   | Sunrisers Hyderabad         | Royal Challengers Bangalore | Royal Challengers Bangalore | field  |
|   | 296   | 2011   | Dharamsala | 2011-05-17 | Kings XI Punjab             | Royal Challengers Bangalore | Kings XI Punjab             | bat    |
|   | 115   | 2008   | Mumbai     | 2008-05-30 | Rajasthan Royals            | Delhi Daredevils            | Delhi Daredevils            | field  |
|   | 7934  | 2018   | Kolkata    | 09/05/18   | Mumbai Indians              | Kolkata Knight Riders       | Kolkata Knight Riders       | field  |
|   | 183   | 2010   | Delhi      | 2010-03-17 | Mumbai Indians              | Delhi Daredevils            | Delhi Daredevils            | field  |

16. Display the 10 matches with the highest win\_by\_wickets values

Limit to 1000 rows

```

46 • SELECT * from matches
47 order by win_by_runs DESC
48 limit 10;
49
50 • SELECT * from matches
51 order by win_by_wickets DESC
52 limit 10;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows: |

|   | id   | season | city       | date       | team1            | team2                       | toss_winner                 | toss_decision | re |
|---|------|--------|------------|------------|------------------|-----------------------------|-----------------------------|---------------|----|
| ▶ | 120  | 2009   | Cape Town  | 2009-04-19 | Kings XI Punjab  | Delhi Daredevils            | Delhi Daredevils            | field         | no |
|   | 377  | 2012   | Jaipur     | 2012-05-20 | Rajasthan Royals | Mumbai Indians              | Rajasthan Royals            | bat           | no |
|   | 3    | 2017   | Rajkot     | 2017-04-07 | Gujarat Lions    | Kolkata Knight Riders       | Kolkata Knight Riders       | field         | no |
|   | 543  | 2015   | Delhi      | 2015-04-26 | Delhi Daredevils | Royal Challengers Bangalore | Royal Challengers Bangalore | field         | no |
|   | 299  | 2011   | Mumbai     | 2011-05-20 | Mumbai Indians   | Rajasthan Royals            | Mumbai Indians              | bat           | no |
|   | 184  | 2010   | Bangalore  | 2010-03-18 | Rajasthan Royals | Royal Challengers Bangalore | Royal Challengers Bangalore | field         | no |
|   | 35   | 2017   | Chandigarh | 2017-04-30 | Delhi Daredevils | Kings XI Punjab             | Kings XI Punjab             | field         | no |
|   | 391  | 2013   | Chandigarh | 2013-04-10 | Kings XI Punjab  | Chennai Super Kings         | Chennai Super Kings         | field         | no |
|   | 7941 | 2018   | Indore     | 14/05/18   | Kings XI Punjab  | Royal Challengers Bangalore | Royal Challengers Bangalore | field         | no |
|   | 72   | 2008   | Mumbai     | 2008-04-27 | Mumbai Indians   | Deccan Chargers             | Deccan Chargers             | field         | no |

Result Grid | Form Editor | Field Types

## Part B — Grouping and Aggregation

17. Count the number of matches played in each season.

The screenshot shows a SQL IDE interface. The query editor contains the following SQL code:

```
49
50 • SELECT * from matches
51   order by win_by_wickets DESC
52   limit 10;
53
54 • SELECT season, COUNT(*) from matches
55   group by season;
```

The 'Result Grid' tab is active, displaying the results of the second query. The table has two columns: 'season' and 'COUNT(\*)'. The data is as follows:

| season | COUNT(*) |
|--------|----------|
| 2017   | 59       |
| 2008   | 58       |
| 2009   | 57       |
| 2010   | 60       |
| 2011   | 73       |
| 2012   | 74       |
| 2013   | 76       |
| 2014   | 60       |
| 2015   | 59       |
| 2016   | 60       |
| 2018   | 60       |
| 2019   | 60       |

18. Count the number of matches played in each city.

The screenshot shows the same SQL IDE interface. The query editor contains the following SQL code:

```
53
54 • SELECT season, COUNT(*) from matches
55   group by season;
56
57 • SELECT city, COUNT(*) from matches
58   group by city;
59
```

The 'Result Grid' tab is active, displaying the results of the second query. The table has two columns: 'city' and 'COUNT(\*)'. The data is as follows:

| city       | COUNT(*) |
|------------|----------|
| Hyderabad  | 64       |
| Pune       | 38       |
| Rajkot     | 10       |
| Indore     | 9        |
| Bangalore  | 66       |
| Mumbai     | 101      |
| Kolkata    | 77       |
| Delhi      | 74       |
| Chandigarh | 46       |
| Kanpur     | 4        |
| Jaipur     | 47       |
| Chennai    | 57       |

19. Count the number of matches played at each venue.

The screenshot shows a SQL IDE interface. The query editor contains the following SQL code:

```
56  
57 • SELECT city, COUNT(*) from matches  
58   group by city;  
59  
60 • SELECT venue, COUNT(*) from matches  
61   group by venue;  
62
```

The result grid below the query shows the following data:

| venue                                              | COUNT(*) |
|----------------------------------------------------|----------|
| Rajiv Gandhi International Stadium, Uppal          | 56       |
| Maharashtra Cricket Association Stadium            | 21       |
| Saurashtra Cricket Association Stadium             | 10       |
| Holkar Cricket Stadium                             | 9        |
| M Chinnaswamy Stadium                              | 73       |
| Wankhede Stadium                                   | 73       |
| Eden Gardens                                       | 77       |
| Feroz Shah Kotla                                   | 67       |
| Punjab Cricket Association IS Bindra Stadium, M... | 14       |
| Green Park                                         | 4        |
| Punjab Cricket Association Stadium, Mohali         | 35       |
| Sawai Mansingh Stadium                             | 47       |

20. Count the number of matches won by each team.

The screenshot shows a SQL IDE interface. The query editor contains the following SQL code:

```
59  
60 • SELECT venue, COUNT(*) from matches  
61   group by venue;  
62  
63 • SELECT winner, COUNT(*) from matches  
64   group by winner;  
65
```

The result grid below the query shows the following data:

| winner                      | COUNT(*) |
|-----------------------------|----------|
| Sunrisers Hyderabad         | 58       |
| Rising Pune Supergiant      | 10       |
| Kolkata Knight Riders       | 92       |
| Kings XI Punjab             | 82       |
| Royal Challengers Bangalore | 84       |
| Mumbai Indians              | 109      |
| Delhi Daredevils            | 67       |
| Gujarat Lions               | 13       |
| Chennai Super Kings         | 100      |
| Rajasthan Royals            | 75       |
| Deccan Chargers             | 29       |
| Pune Warriors               | 12       |

21. Count the number of times each team won the toss.

The screenshot shows a SQL query editor with the following code:

```
62
63 • SELECT winner, COUNT(*) from matches
64   group by winner;
65
66 • SELECT toss_winner, COUNT(*) from matches
67   group by toss_winner;
68
```

Below the query editor is a 'Result Grid' showing the results of the second query. The grid has two columns: 'toss\_winner' and 'COUNT(\*)'. The data is as follows:

| toss_winner                 | COUNT(*) |
|-----------------------------|----------|
| Royal Challengers Bangalore | 81       |
| Rising Pune Supergiant      | 6        |
| Kolkata Knight Riders       | 92       |
| Kings XI Punjab             | 81       |
| Sunrisers Hyderabad         | 46       |
| Mumbai Indians              | 98       |
| Gujarat Lions               | 15       |
| Delhi Daredevils            | 80       |
| Chennai Super Kings         | 89       |
| Rajasthan Royals            | 80       |
| Deccan Chargers             | 43       |
| Kochi Tuskers Kerala        | 8        |

22. Count the number of bat and field toss decisions.

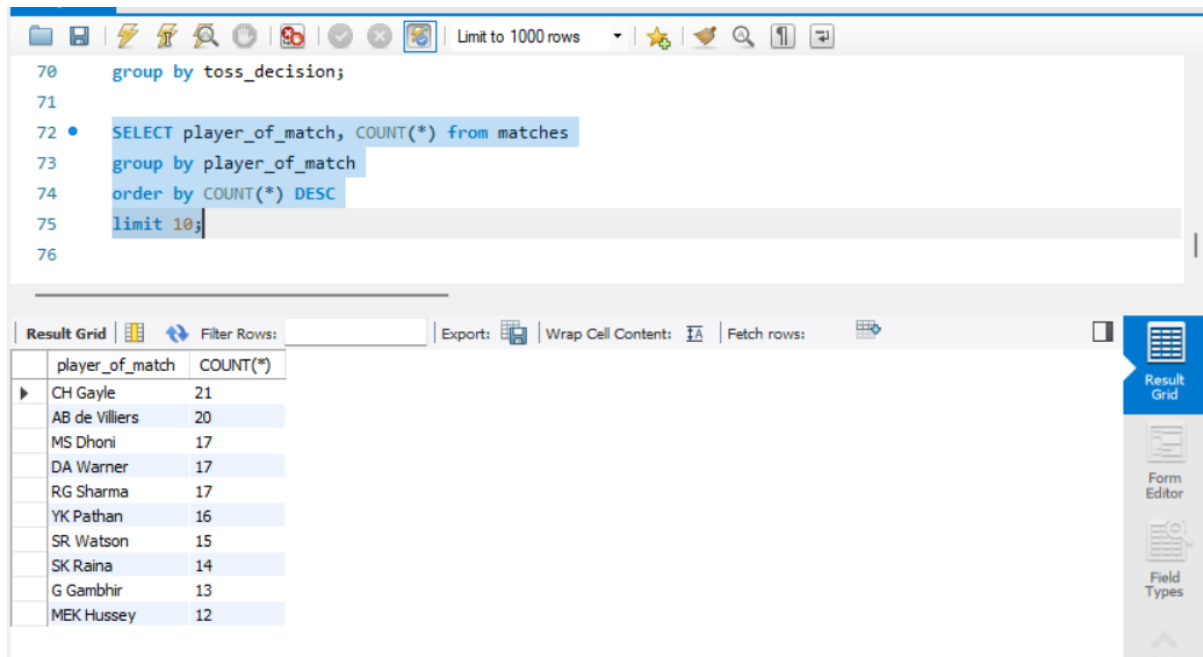
The screenshot shows a SQL query editor with the following code:

```
69 • SELECT toss_decision, COUNT(*) from matches
70   group by toss_decision;
71
```

Below the query editor is a 'Result Grid' showing the results of the query. The grid has two columns: 'toss\_decision' and 'COUNT(\*)'. The data is as follows:

| toss_decision | COUNT(*) |
|---------------|----------|
| field         | 463      |
| bat           | 293      |

23. Find the top 10 players with the most Player of the Match awards.

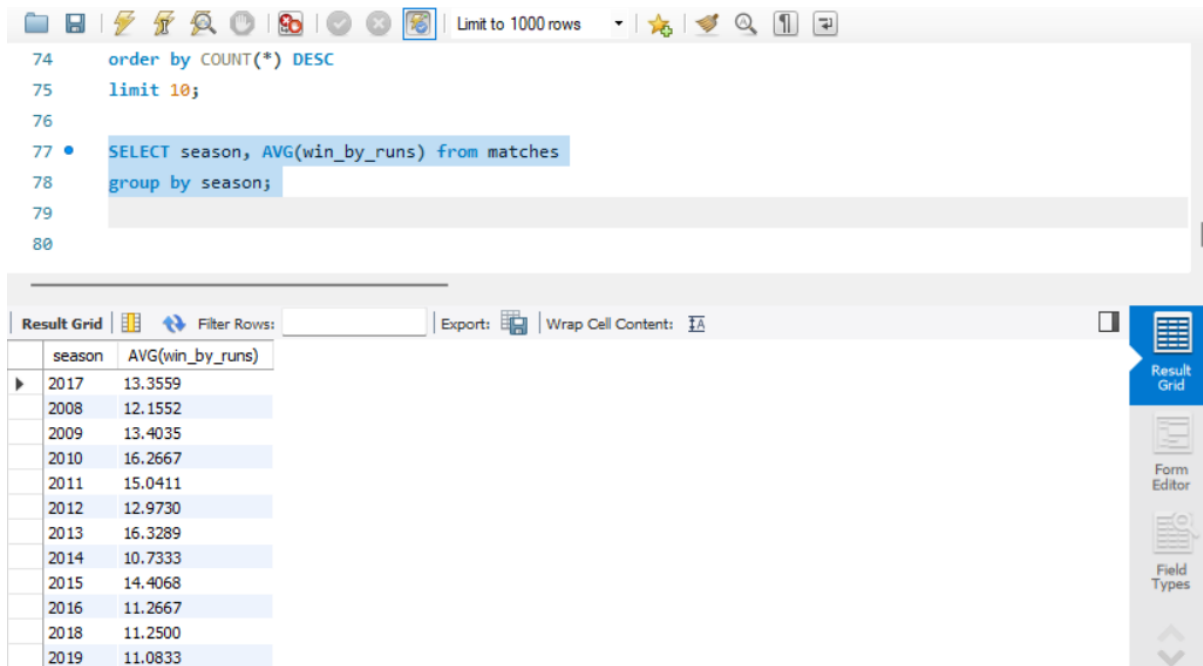


```
70 group by toss_decision;
71
72 • SELECT player_of_match, COUNT(*) from matches
73 group by player_of_match
74 order by COUNT(*) DESC
75 limit 10;
76
```

Result Grid

| player_of_match | COUNT(*) |
|-----------------|----------|
| CH Gayle        | 21       |
| AB de Villiers  | 20       |
| MS Dhoni        | 17       |
| DA Warner       | 17       |
| RG Sharma       | 17       |
| YK Pathan       | 16       |
| SR Watson       | 15       |
| SK Raina        | 14       |
| G Gambhir       | 13       |
| MEK Hussey      | 12       |

24. Find the average win\_by\_runs for each season.

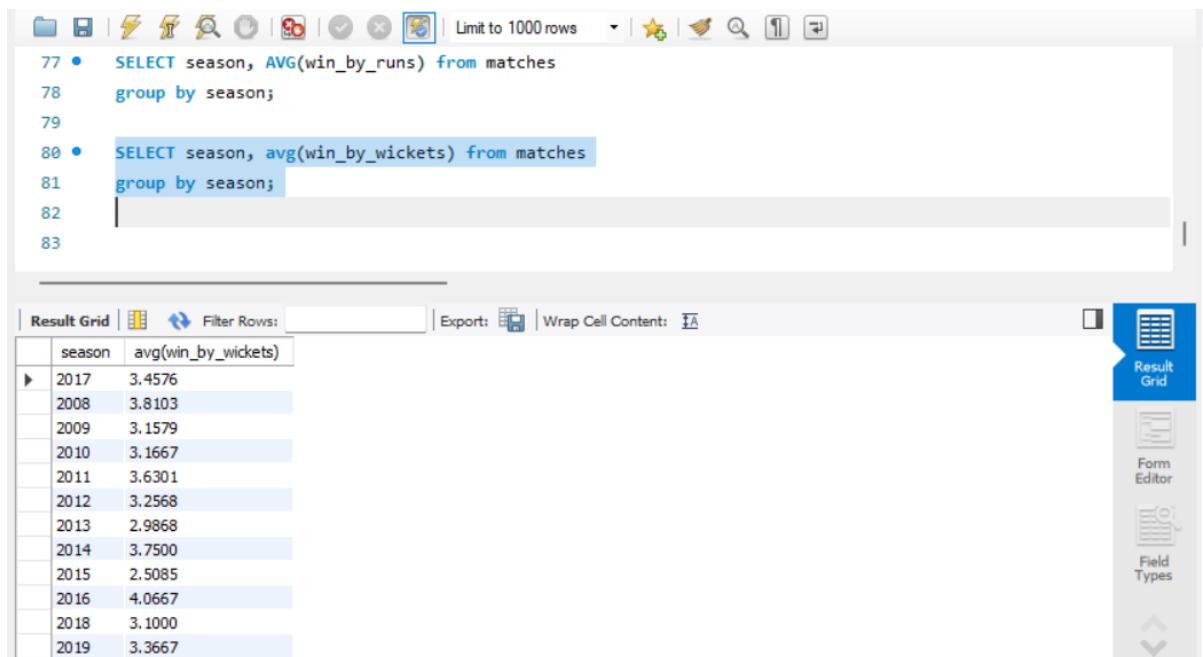


```
74 order by COUNT(*) DESC
75 limit 10;
76
77 • SELECT season, AVG(win_by_runs) from matches
78 group by season;
79
80
```

Result Grid

| season | AVG(win_by_runs) |
|--------|------------------|
| 2017   | 13.3559          |
| 2008   | 12.1552          |
| 2009   | 13.4035          |
| 2010   | 16.2667          |
| 2011   | 15.0411          |
| 2012   | 12.9730          |
| 2013   | 16.3289          |
| 2014   | 10.7333          |
| 2015   | 14.4068          |
| 2016   | 11.2667          |
| 2018   | 11.2500          |
| 2019   | 11.0833          |

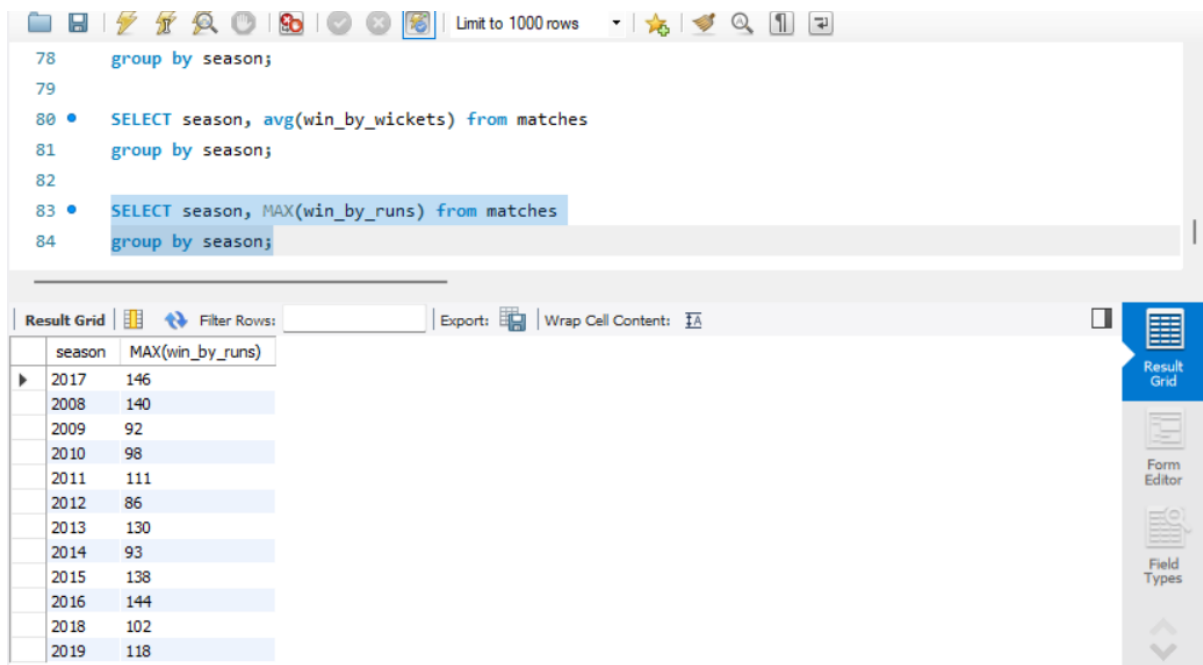
25. Find the average win\_by\_wickets for each season.



```
77 • SELECT season, AVG(win_by_runs) from matches
78   group by season;
79
80 • SELECT season, avg(win_by_wickets) from matches
81   group by season;
82
83
```

| season | avg(win_by_wickets) |
|--------|---------------------|
| 2017   | 3.4576              |
| 2008   | 3.8103              |
| 2009   | 3.1579              |
| 2010   | 3.1667              |
| 2011   | 3.6301              |
| 2012   | 3.2568              |
| 2013   | 2.9868              |
| 2014   | 3.7500              |
| 2015   | 2.5085              |
| 2016   | 4.0667              |
| 2018   | 3.1000              |
| 2019   | 3.3667              |

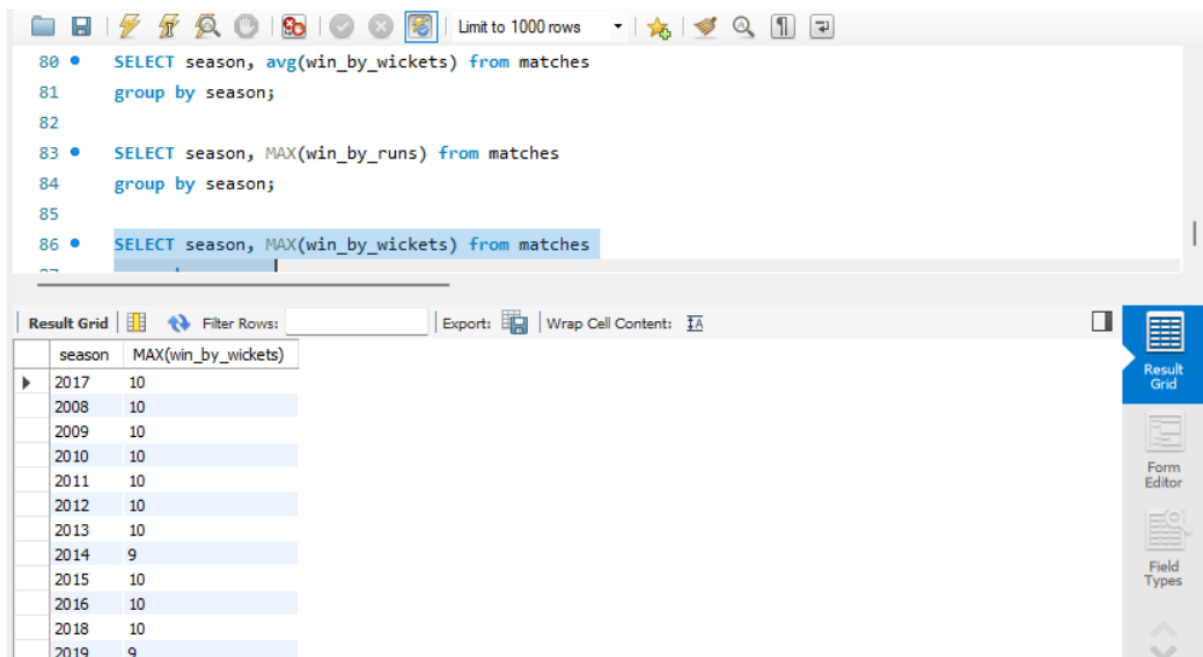
26. Find the maximum win\_by\_runs recorded in each season.



```
78   group by season;
79
80 • SELECT season, avg(win_by_wickets) from matches
81   group by season;
82
83 • SELECT season, MAX(win_by_runs) from matches
84   group by season;
```

| season | MAX(win_by_runs) |
|--------|------------------|
| 2017   | 146              |
| 2008   | 140              |
| 2009   | 92               |
| 2010   | 98               |
| 2011   | 111              |
| 2012   | 86               |
| 2013   | 130              |
| 2014   | 93               |
| 2015   | 138              |
| 2016   | 144              |
| 2018   | 102              |
| 2019   | 118              |

27. Find the maximum win\_by\_wickets recorded in each season.



The screenshot shows a database query editor interface. The top toolbar includes icons for file operations, search, and execution, along with a "Limit to 1000 rows" dropdown. The SQL editor contains three queries:

```
80 • SELECT season, avg(win_by_wickets) from matches
81   group by season;
82
83 • SELECT season, MAX(win_by_runs) from matches
84   group by season;
85
86 • SELECT season, MAX(win_by_wickets) from matches
```

The third query is selected. Below the editor is a toolbar with "Result Grid", "Filter Rows:", "Export:", and "Wrap Cell Content:". The "Result Grid" is active, displaying the following data:

| season | MAX(win_by_wickets) |
|--------|---------------------|
| 2017   | 10                  |
| 2008   | 10                  |
| 2009   | 10                  |
| 2010   | 10                  |
| 2011   | 10                  |
| 2012   | 10                  |
| 2013   | 10                  |
| 2014   | 9                   |
| 2015   | 10                  |
| 2016   | 10                  |
| 2018   | 10                  |
| 2019   | 9                   |

On the right side, there is a vertical toolbar with buttons for "Result Grid", "Form Editor", and "Field Types".

## Part C — Ball-by-Ball Analysis

28. Count the total number of sixes (batsman\_runs = 6) in the deliveries table.

The screenshot shows a SQL IDE interface. The query editor contains the following SQL code:

```
84 group by season;
85
86 • SELECT season, MAX(win_by_wickets) from matches
87 group by season;
88
89 • SELECT COUNT(*) from deliveries
90 where batsman_runs = 6;
```

The result grid at the bottom shows a single row with the value 2624.

| COUNT(*) |
|----------|
| 2624     |

The right sidebar contains buttons for Result Grid, Form Editor, and Field Types.

29. Count the total number of fours (batsman\_runs = 4) in the deliveries table

The screenshot shows a SQL IDE interface. The query editor contains the following SQL code:

```
87 group by season;
88
89 • SELECT COUNT(*) from deliveries
90 where batsman_runs = 6;
91
92 • SELECT COUNT(*) from deliveries
93 where batsman_runs = 4;
```

The result grid at the bottom shows a single row with the value 6991.

| COUNT(*) |
|----------|
| 6991     |

The right sidebar contains buttons for Result Grid, Form Editor, and Field Types.

30. Find the total number of wide runs in the deliveries table.

The screenshot shows a SQL IDE interface. The query editor contains the following SQL code:

```
89 • SELECT COUNT(*) from deliveries
90   where batsman_runs = 6;
91
92 • SELECT COUNT(*) from deliveries
93   where batsman_runs = 4;
94
95 • SELECT sum(wide_runs) from deliveries;
```

The third query is selected and highlighted. Below the editor, the 'Result Grid' tab is active, displaying the result of the selected query:

| sum(wide_runs) |
|----------------|
| 2446           |

The interface includes a toolbar at the top with various icons and a 'Limit to 1000 rows' dropdown. On the right side, there are buttons for 'Result Grid', 'Form Editor', and 'Field Types'.

31. Find the total number of no-ball runs in the deliveries table.

The screenshot shows a SQL IDE interface. The query editor contains the following SQL code:

```
91
92 • SELECT COUNT(*) from deliveries
93   where batsman_runs = 4;
94
95 • SELECT sum(wide_runs) from deliveries;
96
97 • SELECT sum(noball_runs) from deliveries;
```

The third query is selected and highlighted. Below the editor, the 'Result Grid' tab is active, displaying the result of the selected query:

| sum(noball_runs) |
|------------------|
| 309              |

The interface includes a toolbar at the top with various icons and a 'Limit to 1000 rows' dropdown. On the right side, there are buttons for 'Result Grid', 'Form Editor', and 'Field Types'.

32. Find the total batsman runs scored by each batsman and show the top 10 run scorers.

The screenshot shows a database query editor with a SQL query entered in the editor pane. The query is:

```
SELECT sum(noball_runs) from deliveries;  
  
SELECT batsman, sum(batsman_runs) from deliveries  
group by batsman  
order by sum(batsman_runs) DESC  
limit 10;
```

The result grid below the query shows the following data:

| batsman      | sum(batsman_runs) |
|--------------|-------------------|
| SK Raina     | 1942              |
| G Gambhir    | 1738              |
| RG Sharma    | 1637              |
| MS Dhoni     | 1434              |
| SR Tendulkar | 1411              |
| RV Uthappa   | 1359              |
| JH Kallis    | 1329              |
| AC Gilchrist | 1329              |
| Yuvraj Singh | 1245              |
| DA Warner    | 1238              |

33. Find the number of sixes hit by each batsman and show the top 10 six hitters.

The screenshot shows a database query editor with a SQL query entered in the editor pane. The query is:

```
SELECT batsman, COUNT(*) from deliveries  
where batsman_runs = 6  
group by batsman  
order by COUNT(*) DESC  
limit 10;
```

The result grid below the query shows the following data:

| batsman      | COUNT(*) |
|--------------|----------|
| SK Raina     | 77       |
| YK Pathan    | 71       |
| AC Gilchrist | 68       |
| RG Sharma    | 66       |
| RV Uthappa   | 65       |
| Yuvraj Singh | 63       |
| BB McCullum  | 55       |
| MS Dhoni     | 53       |
| DA Warner    | 49       |
| ML Hayden    | 44       |

34. Find the number of fours hit by each batsman and show the top 10 four hitters.

```
109
110 • SELECT batsman, COUNT(*) from deliveries
111     where batsman_runs = 4
112     group by batsman
113     order by COUNT(*) DESC
114     limit 10;
115
```

Result Grid

| batsman       | COUNT(*) |
|---------------|----------|
| G Gambhir     | 212      |
| SR Tendulkar  | 176      |
| SK Raina      | 172      |
| JH Kallis     | 149      |
| AC Gilchrist  | 147      |
| RG Sharma     | 139      |
| KC Sangakkara | 135      |
| V Sehwag      | 129      |
| S Dhawan      | 126      |
| PA Patel      | 125      |

35. Find the total runs scored by each batting team.

```
112     group by batsman
113     order by COUNT(*) DESC
114     limit 10;
115
116 • SELECT batting_team, sum(total_runs) from deliveries
117     group by batting_team;
118
```

Result Grid

| batting_team                | sum(total_runs) |
|-----------------------------|-----------------|
| Sunrisers Hyderabad         | 2349            |
| Royal Challengers Bangalore | 9115            |
| Mumbai Indians              | 10227           |
| Rising Pune Supergiant      | 2470            |
| Gujarat Lions               | 2412            |
| Kolkata Knight Riders       | 9038            |
| Kings XI Punjab             | 9553            |
| Delhi Daredevils            | 9211            |
| Chennai Super Kings         | 8166            |
| Rajasthan Royals            | 7166            |
| Deccan Chargers             | 7810            |
| Kochi Tuskers Kerala        | 760             |

36. Find the total extra runs received by each batting team.

```
115
116 • SELECT batting_team, sum(total_runs) from deliveries
117     group by batting_team;
118
119 • SELECT batting_team, sum(extra_runs) from deliveries
120     group by batting_team;
121
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

| batting_team                | sum(extra_runs) |
|-----------------------------|-----------------|
| Sunrisers Hyderabad         | 87              |
| Royal Challengers Bangalore | 491             |
| Mumbai Indians              | 721             |
| Rising Pune Supergiant      | 100             |
| Gujarat Lions               | 101             |
| Kolkata Knight Riders       | 582             |
| Kings XI Punjab             | 557             |
| Delhi Daredevils            | 479             |
| Chennai Super Kings         | 489             |
| Rajasthan Royals            | 423             |
| Deccan Chargers             | 397             |
| Kochi Tuskers Kerala        | 44              |

Result Grid  
Form Editor  
Field Types

37. Count the number of deliveries bowled by each bowler and show the top 10.

```
120     group by batting_team;
121
122 • SELECT bowler, COUNT(*) from deliveries
123     group by bowler
124     order by COUNT(*) DESC
125     limit 10;
126
```

Limit to 1000 rows | | | | |

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows: |

| bowler          | COUNT(*) |
|-----------------|----------|
| RP Singh        | 1086     |
| P Kumar         | 1074     |
| A Mishra        | 1072     |
| SK Warne        | 1058     |
| M Muralitharan  | 1052     |
| SL Malinga      | 1051     |
| PP Chawla       | 1037     |
| IK Pathan       | 1035     |
| Z Khan          | 1033     |
| Harbhajan Singh | 1005     |

Result 42 | Read Only

38. Find the total runs conceded by each bowler using total\_runs and show the top 10 highest totals.

The screenshot shows a SQL query editor with the following code:

```
125 limit 10;
126
127 • SELECT bowler, sum(total_runs) from deliveries
128 group by bowler
129 order by COUNT(*) DESC
130 limit 10;
131
```

Below the editor is a result grid showing the top 10 bowlers by total runs conceded:

| bowler          | sum(total_runs) |
|-----------------|-----------------|
| RP Singh        | 1409            |
| P Kumar         | 1443            |
| A Mishra        | 1304            |
| SK Warne        | 1301            |
| M Muralitharan  | 1130            |
| SL Malinga      | 1218            |
| PP Chawla       | 1351            |
| IK Pathan       | 1324            |
| Z Khan          | 1368            |
| Harbhajan Singh | 1103            |

The interface includes a toolbar at the top with icons for file operations, a 'Limit to 1000 rows' dropdown, and a sidebar on the right with 'Result Grid', 'Form Editor', and 'Field Types' buttons.

39. Count the number of recorded dismissals for each bowler, excluding run out, and show the top 10.

The screenshot shows a SQL query editor with the following code:

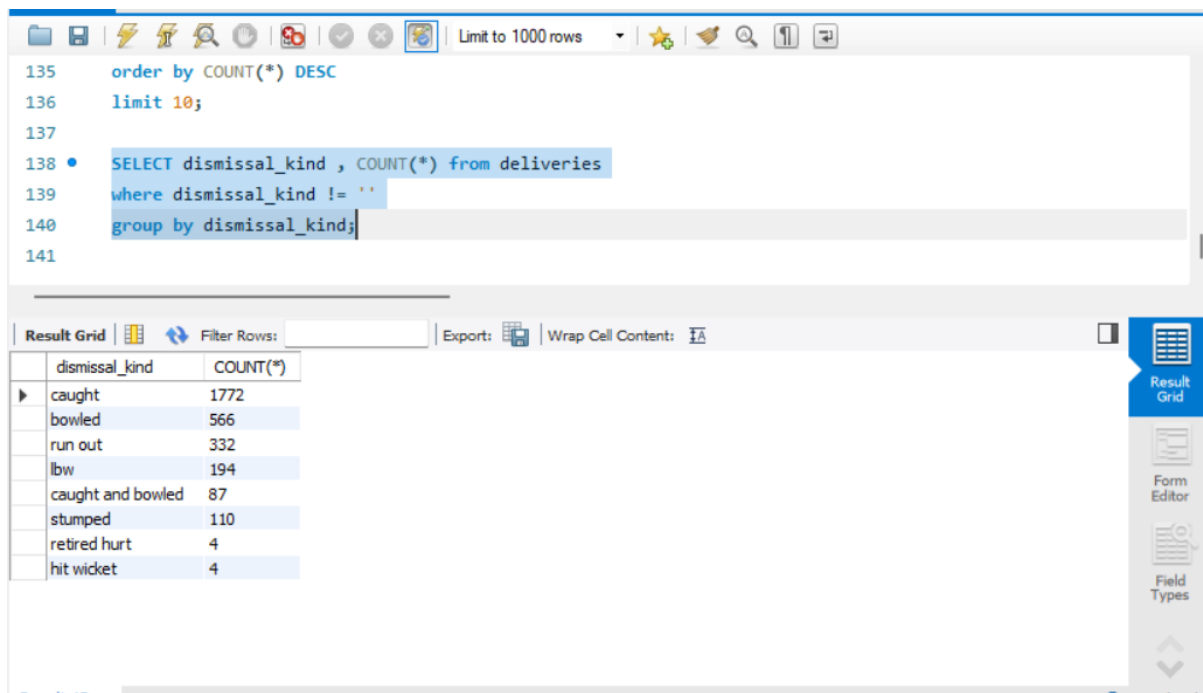
```
131
132 • SELECT bowler, COUNT(*) from deliveries
133 where dismissal_kind != 'run out' AND dismissal_kind != ''
134 group by bowler
135 order by COUNT(*) DESC
136 limit 10;
137
```

Below the editor is a result grid showing the top 10 bowlers by the number of dismissals (excluding run outs):

| bowler          | COUNT(*) |
|-----------------|----------|
| A Mishra        | 58       |
| SL Malinga      | 57       |
| RP Singh        | 54       |
| PP Ojha         | 51       |
| Z Khan          | 50       |
| PP Chawla       | 50       |
| SK Warne        | 50       |
| IK Pathan       | 49       |
| Harbhajan Singh | 49       |
| A Kumble        | 45       |

The interface includes a toolbar at the top with icons for file operations, a 'Limit to 1000 rows' dropdown, and a sidebar on the right with 'Result Grid', 'Form Editor', and 'Field Types' buttons.

40. Count how many dismissals occurred for each dismissal\_kind.



The screenshot shows a SQL IDE interface. The top toolbar includes icons for file operations, execution, and a 'Limit to 1000 rows' dropdown. The SQL editor contains the following code:

```
135 order by COUNT(*) DESC
136 limit 10;
137
138 • SELECT dismissal_kind , COUNT(*) from deliveries
139 where dismissal_kind != ''
140 group by dismissal_kind;
141
```

Below the editor is the 'Result Grid' section, which includes a 'Filter Rows' input, an 'Export' button, and a 'Wrap Cell Content' checkbox. The results are displayed in a table:

| dismissal_kind    | COUNT(*) |
|-------------------|----------|
| caught            | 1772     |
| bowled            | 566      |
| run out           | 332      |
| lbw               | 194      |
| caught and bowled | 87       |
| stumped           | 110      |
| retired hurt      | 4        |
| hit wicket        | 4        |

On the right side of the interface, there is a vertical toolbar with icons for 'Result Grid', 'Form Editor', and 'Field Types'.

## Part D — Analysis Using Both Tables

41. Join matches and deliveries using matches.id = deliveries.match\_id and display 20 joined records.

The screenshot shows a SQL query editor with the following code:

```
140 group by dismissal_kind;
141
142 • SELECT * from matches
143 JOIN deliveries ON matches.id = deliveries.match_id
144 limit 20;
145
146
```

Below the editor is the 'Result Grid' showing 20 rows of data. The columns are: id, season, city, date, team1, team2, toss\_winner, and toss\_decision. The data shows matches from the 2017 season in Hyderabad, all involving Sunrisers Hyderabad and Royal Challengers Bangalore, with toss decisions being 'field'.

| id | season | city      | date       | team1               | team2                       | toss_winner                 | toss_decision |
|----|--------|-----------|------------|---------------------|-----------------------------|-----------------------------|---------------|
| 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad | Royal Challengers Bangalore | Royal Challengers Bangalore | field         |
| 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad | Royal Challengers Bangalore | Royal Challengers Bangalore | field         |
| 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad | Royal Challengers Bangalore | Royal Challengers Bangalore | field         |
| 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad | Royal Challengers Bangalore | Royal Challengers Bangalore | field         |
| 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad | Royal Challengers Bangalore | Royal Challengers Bangalore | field         |
| 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad | Royal Challengers Bangalore | Royal Challengers Bangalore | field         |
| 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad | Royal Challengers Bangalore | Royal Challengers Bangalore | field         |
| 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad | Royal Challengers Bangalore | Royal Challengers Bangalore | field         |
| 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad | Royal Challengers Bangalore | Royal Challengers Bangalore | field         |
| 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad | Royal Challengers Bangalore | Royal Challengers Bangalore | field         |
| 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad | Royal Challengers Bangalore | Royal Challengers Bangalore | field         |
| 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad | Royal Challengers Bangalore | Royal Challengers Bangalore | field         |
| 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad | Royal Challengers Bangalore | Royal Challengers Bangalore | field         |
| 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad | Royal Challengers Bangalore | Royal Challengers Bangalore | field         |
| 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad | Royal Challengers Bangalore | Royal Challengers Bangalore | field         |
| 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad | Royal Challengers Bangalore | Royal Challengers Bangalore | field         |
| 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad | Royal Challengers Bangalore | Royal Challengers Bangalore | field         |
| 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad | Royal Challengers Bangalore | Royal Challengers Bangalore | field         |
| 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad | Royal Challengers Bangalore | Royal Challengers Bangalore | field         |
| 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad | Royal Challengers Bangalore | Royal Challengers Bangalore | field         |
| 1  | 2017   | Hyderabad | 2017-04-05 | Sunrisers Hyderabad | Royal Challengers Bangalore | Royal Challengers Bangalore | field         |

42. Using the joined tables, find the total runs scored in each IPL season.

The screenshot shows a SQL query editor with the following code:

```
144 limit 20;
145
146 • SELECT matches.season, sum(deliveries.total_runs) from matches
147 JOIN deliveries ON matches.id = deliveries.match_id
148 group by matches.season;
149
150
```

Below the editor is the 'Result Grid' showing 5 rows of data. The columns are: season and sum(deliveries.total\_runs). The data shows the total runs scored in each IPL season from 2017 to 2011.

| season | sum(deliveries.total_runs) |
|--------|----------------------------|
| 2017   | 18786                      |
| 2008   | 17937                      |
| 2009   | 16353                      |
| 2010   | 18883                      |
| 2011   | 6887                       |

43. Using the joined tables, find the total number of sixes in each season.

The screenshot shows a database query editor with a SQL query to count the number of sixes in each season. The query is as follows:

```
149
150 • SELECT matches.season, COUNT(*) from matches
151 JOIN deliveries ON matches.id = deliveries.match_id
152 where batsman_runs = 6
153 group by matches.season;
154
155
```

Below the query editor, the 'Result Grid' is displayed, showing the results of the query. The grid has two columns: 'season' and 'COUNT(\*)'. The results are as follows:

| season | COUNT(*) |
|--------|----------|
| 2017   | 706      |
| 2008   | 623      |
| 2009   | 508      |
| 2010   | 587      |
| 2011   | 200      |

The interface includes a toolbar at the top with various icons and a 'Limit to 1000 rows' dropdown. On the right side, there is a sidebar with buttons for 'Result Grid', 'Form Editor', and 'Field Types'.

44. Using the joined tables, find the total number of fours in each season.

The screenshot shows a database query editor with a SQL query to count the number of fours in each season. The query is as follows:

```
154
155 • SELECT matches.season, COUNT(*) from matches
156 JOIN deliveries ON matches.id = deliveries.match_id
157 where batsman_runs =4
158 group by matches.season;
159
160
```

Below the query editor, the 'Result Grid' is displayed, showing the results of the query. The grid has two columns: 'season' and 'COUNT(\*)'. The results are as follows:

| season | COUNT(*) |
|--------|----------|
| 2017   | 1612     |
| 2008   | 1703     |
| 2009   | 1321     |
| 2010   | 1709     |
| 2011   | 646      |

The interface includes a toolbar at the top with various icons and a 'Limit to 1000 rows' dropdown. On the right side, there is a sidebar with buttons for 'Result Grid', 'Form Editor', and 'Field Types'.

45. Using the joined tables, find the total extra runs in each season.

The screenshot shows a SQL query editor with the following code:

```
158 group by matches.season;
159
160 • SELECT matches.season , sum(deliveries.extra_runs) from matches
161 JOIN deliveries ON matches.id = deliveries.match_id
162 group by matches.season;
163
164
```

Below the query editor, the 'Result Grid' is displayed, showing the results of the query:

| season | sum(deliveries.extra_runs) |
|--------|----------------------------|
| 2017   | 866                        |
| 2008   | 1128                       |
| 2009   | 977                        |
| 2010   | 1129                       |
| 2011   | 406                        |

The interface includes a toolbar at the top with icons for file operations, a 'Limit to 1000 rows' dropdown, and a 'Filter Rows' input field. On the right side, there are buttons for 'Result Grid', 'Form Editor', and 'Field Types'.

46. Using the joined tables, find total runs scored by each batting team in each season.

The screenshot shows a SQL query editor with the following code:

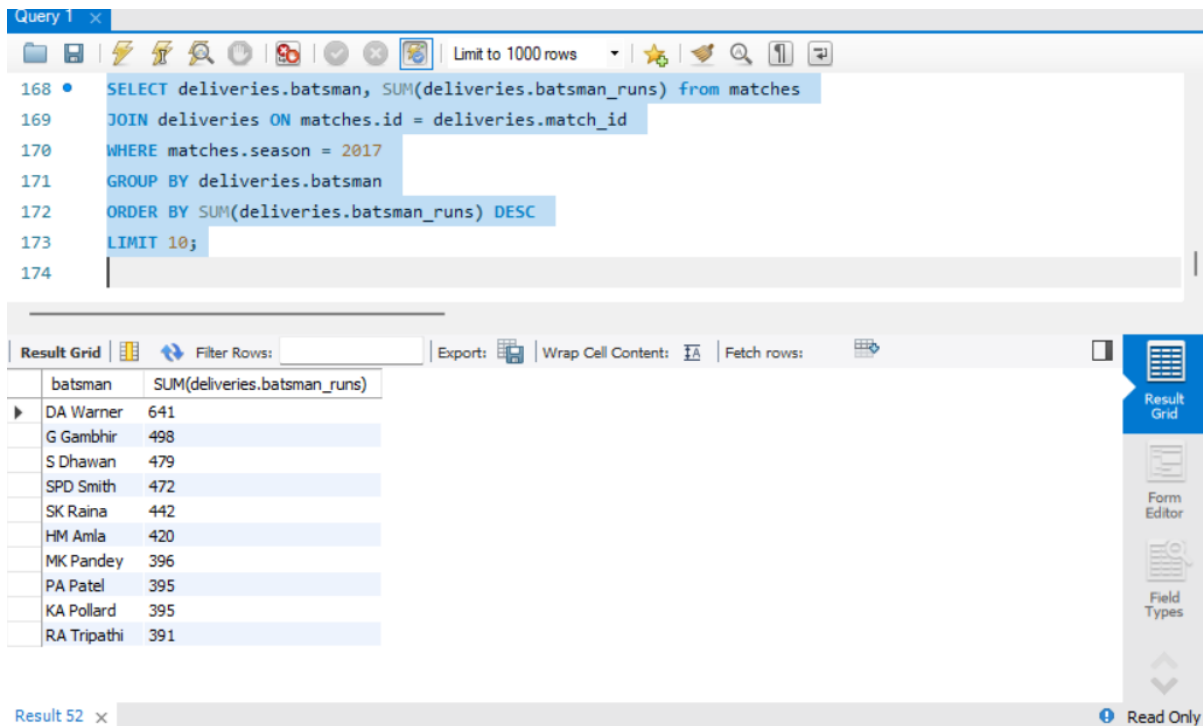
```
162 group by matches.season;
163
164 • SELECT matches.season, deliveries.batting_team, sum(deliveries.total_runs) from matches
165 JOIN deliveries ON matches.id = deliveries.match_id
166 GROUP BY matches.season, deliveries.batting_team;
167
168
```

Below the query editor, the 'Result Grid' is displayed, showing the results of the query:

| season | batting_team                | sum(deliveries.total_runs) |
|--------|-----------------------------|----------------------------|
| 2017   | Sunrisers Hyderabad         | 2349                       |
| 2017   | Royal Challengers Bangalore | 1845                       |
| 2017   | Mumbai Indians              | 2800                       |
| 2017   | Rising Pune Supergiant      | 2470                       |
| 2017   | Gujarat Lions               | 2412                       |
| 2017   | Kolkata Knight Riders       | 2484                       |
| 2017   | Kings XI Punjab             | 2207                       |
| 2017   | Delhi Daredevils            | 2219                       |
| 2008   | Kolkata Knight Riders       | 1942                       |
| 2008   | Royal Challengers Bangalore | 1983                       |
| 2008   | Chennai Super Kings         | 2520                       |
| 2008   | Kings XI Punjab             | 2464                       |

The interface includes a toolbar at the top with icons for file operations, a 'Limit to 1000 rows' dropdown, and a 'Filter Rows' input field. On the right side, there are buttons for 'Result Grid', 'Form Editor', and 'Field Types'. At the bottom left, it says 'Result 51 x' and at the bottom right, there is a 'Read Only' status indicator.

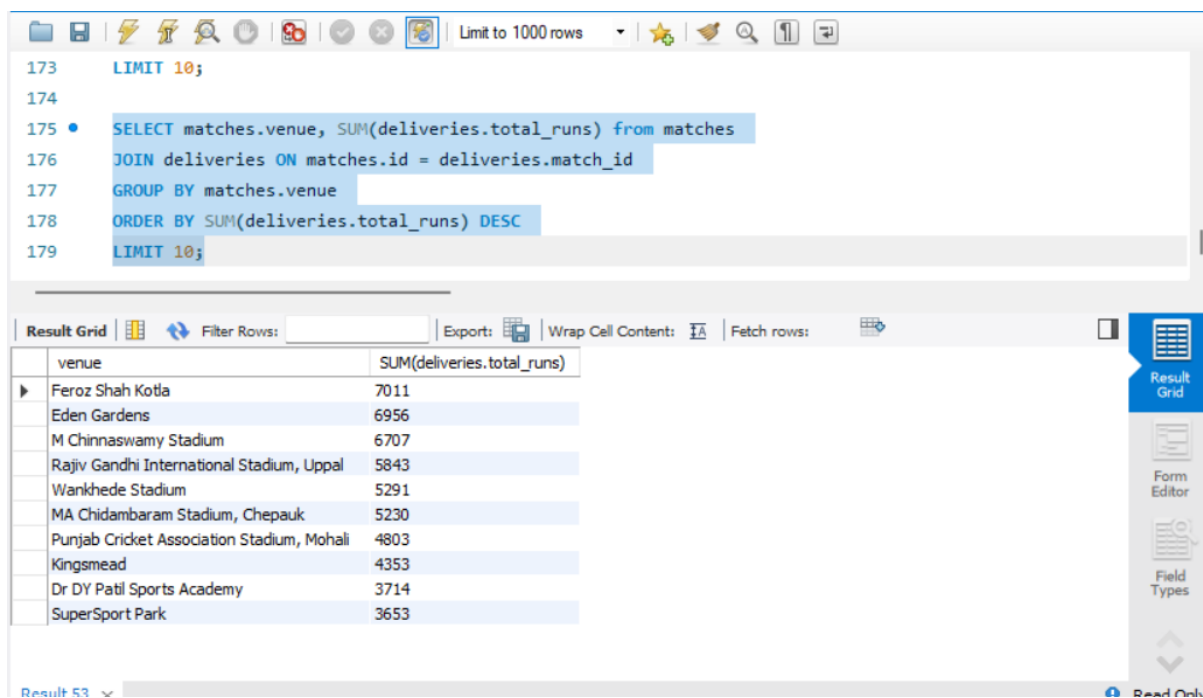
47. Using the joined tables, find the top 10 run scorers in the 2017 season.



```
168 • SELECT deliveries.batsman, SUM(deliveries.batsman_runs) from matches
169 JOIN deliveries ON matches.id = deliveries.match_id
170 WHERE matches.season = 2017
171 GROUP BY deliveries.batsman
172 ORDER BY SUM(deliveries.batsman_runs) DESC
173 LIMIT 10;
174
```

| batsman     | SUM(deliveries.batsman_runs) |
|-------------|------------------------------|
| DA Warner   | 641                          |
| G Gambhir   | 498                          |
| S Dhawan    | 479                          |
| SPD Smith   | 472                          |
| SK Raina    | 442                          |
| HM Amla     | 420                          |
| MK Pandey   | 396                          |
| PA Patel    | 395                          |
| KA Pollard  | 395                          |
| RA Tripathi | 391                          |

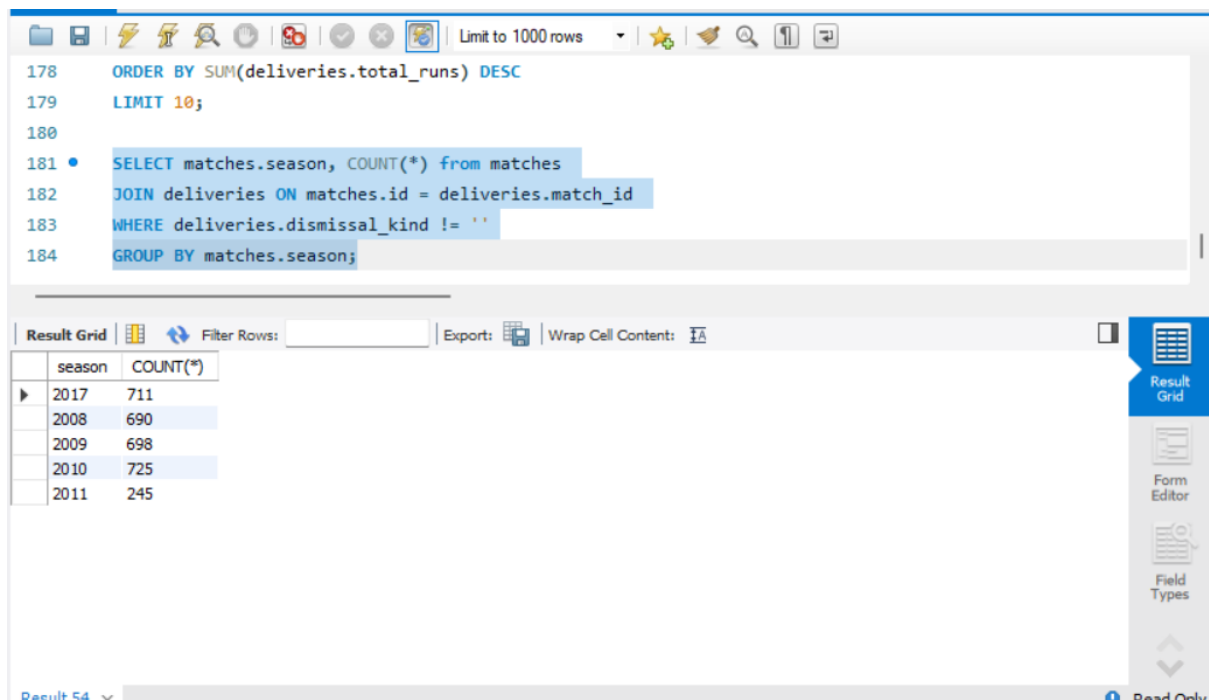
48. Using the joined tables, find the top 10 venues by total runs scored.



```
173 LIMIT 10;
174
175 • SELECT matches.venue, SUM(deliveries.total_runs) from matches
176 JOIN deliveries ON matches.id = deliveries.match_id
177 GROUP BY matches.venue
178 ORDER BY SUM(deliveries.total_runs) DESC
179 LIMIT 10;
```

| venue                                      | SUM(deliveries.total_runs) |
|--------------------------------------------|----------------------------|
| Feroz Shah Kotla                           | 7011                       |
| Eden Gardens                               | 6956                       |
| M Chinnaswamy Stadium                      | 6707                       |
| Rajiv Gandhi International Stadium, Uppal  | 5843                       |
| Wankhede Stadium                           | 5291                       |
| MA Chidambaram Stadium, Chepauk            | 5230                       |
| Punjab Cricket Association Stadium, Mohali | 4803                       |
| Kingsmead                                  | 4353                       |
| Dr DY Patil Sports Academy                 | 3714                       |
| SuperSport Park                            | 3653                       |

49. Using the joined tables, find the number of recorded dismissals in each season.



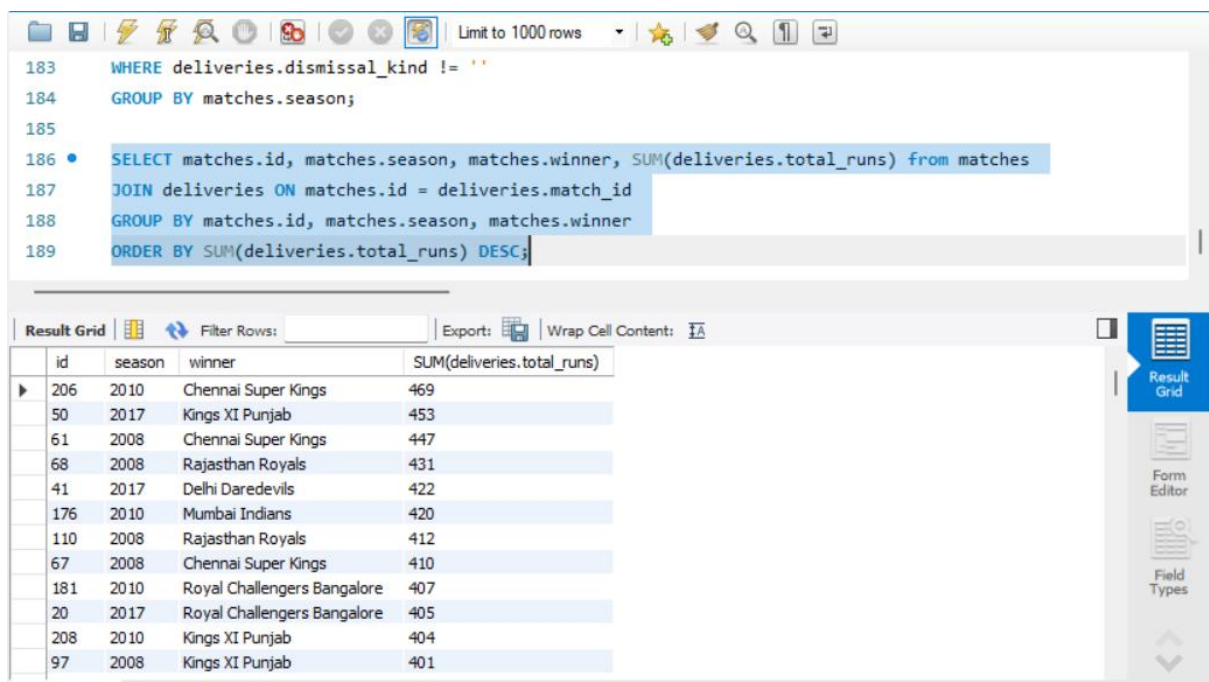
```

178 ORDER BY SUM(deliveries.total_runs) DESC
179 LIMIT 10;
180
181 • SELECT matches.season, COUNT(*) from matches
182 JOIN deliveries ON matches.id = deliveries.match_id
183 WHERE deliveries.dismissal_kind != ''
184 GROUP BY matches.season;

```

| season | COUNT(*) |
|--------|----------|
| 2017   | 711      |
| 2008   | 690      |
| 2009   | 698      |
| 2010   | 725      |
| 2011   | 245      |

50. Using the joined tables, show each match ID, season, winner, and total runs scored in that match, ordered from highest match total to lowest.



```

183 WHERE deliveries.dismissal_kind != ''
184 GROUP BY matches.season;
185
186 • SELECT matches.id, matches.season, matches.winner, SUM(deliveries.total_runs) from matches
187 JOIN deliveries ON matches.id = deliveries.match_id
188 GROUP BY matches.id, matches.season, matches.winner
189 ORDER BY SUM(deliveries.total_runs) DESC;

```

| id  | season | winner                      | SUM(deliveries.total_runs) |
|-----|--------|-----------------------------|----------------------------|
| 206 | 2010   | Chennai Super Kings         | 469                        |
| 50  | 2017   | Kings XI Punjab             | 453                        |
| 61  | 2008   | Chennai Super Kings         | 447                        |
| 68  | 2008   | Rajasthan Royals            | 431                        |
| 41  | 2017   | Delhi Daredevils            | 422                        |
| 176 | 2010   | Mumbai Indians              | 420                        |
| 110 | 2008   | Rajasthan Royals            | 412                        |
| 67  | 2008   | Chennai Super Kings         | 410                        |
| 181 | 2010   | Royal Challengers Bangalore | 407                        |
| 20  | 2017   | Royal Challengers Bangalore | 405                        |
| 208 | 2010   | Kings XI Punjab             | 404                        |
| 97  | 2008   | Kings XI Punjab             | 401                        |